

Project Title: Ski Resort Gear Rental System
Course: CS340
Group: 80
Team Members: Joshua Cabalka, Jack Boland, Philip Gadsden
Project Step 2: Draft — Normalized Schema + DDL + Sample Data
Date: 02/05/2026

a) Fixes Based on Feedback from Step 1

Feedback on the Step 1 submission was mainly positive, with reviewers agreeing that the system design, entities, relationships, and ERD were clear and aligned with requirements. Only minor issues were identified and addressed.

Reviewers noted that the overview could better illustrate system scale. The overview was updated to more explicitly describe operational scope, including individual gear tracking, multi-item rental orders, historical checkout and return records, and independent maintenance workflows.

An error was identified in the database outline where `customer_id` in the `rental_orders` table incorrectly referenced the `employees` table. This was corrected to reference `customers(customer_id)`, aligning the outline with the ERD and business logic.

A minor naming inconsistency was identified in the `rental_order_items` table, where the foreign key name did not match the rental order primary key naming convention. Fixed foreign key reference so `rental_orders.customer_id` correctly references `customers(customer_id)`.

Clarified meaning of `employees.is_active` (employee is currently employed/authorized to create orders and service tickets).

No other changes were required, as reviewers confirmed that entity modeling, relationships, naming conventions, and the ERD were correct.

b) Project Overview and Database Outline

This project models an admin-facing ski resort gear rental system designed to manage an equipment inventory of approximately 300 individual gear items across various categories, along with an average of 80 rental transactions per week and 20 maintenance service tickets per week.

The system tracks individual gear items to allow staff to monitor availability, condition, rental history, and service history at the item level. Rental activity is recorded through rental orders created by employees for customers, with each rental order capable of containing multiple gear items. A typical rental order contains 2–4 gear items. Historical rental data is preserved using timestamps that record when items are checked out and returned.

Maintenance and repair work is tracked independently using service tickets, which document service procedures performed on one or more gear items. Each service ticket includes timestamps for work start and completion, allowing the system to maintain a complete maintenance history separate from rental activity. Together, these components support both daily operational needs and long-term inventory and service tracking for the ski resort.

DATABASE OUTLINE

1. **customers:** Records customers who rent gear from the ski resort.

Attributes

- `customer_id`: INT, AUTO_INCREMENT, PRIMARY KEY, NOT NULL
- `first_name`: VARCHAR(100), NOT NULL
- `last_name`: VARCHAR(100), NOT NULL
- `phone`: VARCHAR(20), NULL
- `email`: VARCHAR(255), NULL

Relationships

- One-to-many (1:M) relationship with `rental_orders`

2. **employees:** Records employees who create rental orders and service tickets.

Attributes

- `employee_id`: INT, AUTO_INCREMENT, PRIMARY KEY, NOT NULL
- `first_name`: VARCHAR(100), NOT NULL
- `last_name`: VARCHAR(100), NOT NULL
- `role`: VARCHAR(50), NOT NULL
- `is_active`: BOOLEAN, NOT NULL, DEFAULT TRUE
- Indicates whether the employee is currently employed and authorized to create rental orders and service tickets

Relationships

- One-to-many (1:M) relationship with `rental_orders`
- One-to-many (1:M) relationship with `service_tickets`

3. **gear_items:** Records each physical piece of ski or snowboard-related equipment.

Attributes

- `gear_item_id`: INT, AUTO_INCREMENT, PRIMARY KEY, NOT NULL
- `category`: ENUM('snowboard', 'boots', 'helmet', 'other'), NOT NULL
- `brand`: VARCHAR(100), NOT NULL
- `model`: VARCHAR(100), NOT NULL
- `serial_number`: VARCHAR(100), UNIQUE, NOT NULL
- `size`: VARCHAR(20), NOT NULL
- `condition_grade`: ENUM('new', 'good', 'fair', 'needs_repair'), NOT NULL
- `status`: ENUM('available', 'rented', 'in_service', 'retired'), NOT NULL
- `acquired_at`: DATETIME, NOT NULL

Relationships

- Many-to-many (M:M) relationship with `rental_orders`
- Many-to-many (M:M) relationship with `service_tickets`

4. **rental_orders:** Records each rental checkout transaction.

Attributes

- `rental_order_id`: INT, AUTO_INCREMENT, PRIMARY KEY, NOT NULL
- `customer_id`: INT, NOT NULL, FOREIGN KEY → `customers(customer_id)`
- `created_by_employee_id`: INT, NOT NULL, FOREIGN KEY → `employees(employee_id)`

- created_at: DATETIME, NOT NULL

Relationships

- Many-to-one (M:1) relationship with customers
- Many-to-one (M:1) relationship with employees
- Many-to-many (M:M) relationship with gear_items (via rental_order_items)

5. **service_tickets**: Records maintenance and repair work performed on gear items.

Attributes

- service_ticket_id: INT, AUTO_INCREMENT, PRIMARY KEY, NOT NULL
- opened_by_employee_id: INT, NOT NULL, FOREIGN KEY → employees(employee_id)
- status: ENUM('open', 'in_progress', 'completed', 'canceled'), NOT NULL
- created_at: DATETIME, NOT NULL

Relationships

- Many-to-one (M:1) relationship with employees
- Many-to-many (M:M) relationship with gear_items (via service_ticket_items)

INTERSECTION TABLES

6. **rental_order_items**: Implements the many-to-many relationship between rental_orders and gear_items.

Attributes

- rental_order_id: INT, NOT NULL, FOREIGN KEY → rental_orders(rental_order_id)
- gear_item_id: INT, NOT NULL, FOREIGN KEY → gear_items(gear_item_id)
- checked_out_at: DATETIME, NOT NULL
- due_at: DATETIME, NOT NULL
- returned_at: DATETIME, NULL

Primary Key

- (rental_order_id, gear_item_id)

Business Rule

- A gear item may only appear in one active rental at a time. This is enforced by checking that gear_items.status = 'available' prior to checkout and updating the status to 'rented' while returned_at is NULL.

7. **service_ticket_items**: Implements the many-to-many relationship between service_tickets and gear_items.

Attributes

- service_ticket_id: INT, NOT NULL, FOREIGN KEY → service_tickets(service_ticket_id)
- gear_item_id: INT, NOT NULL, FOREIGN KEY → gear_items(gear_item_id)
- service_type: ENUM('wax', 'tune', 'repair', 'binding_adjust', 'inspect'), NOT NULL
- started_at: DATETIME, NULL
- completed_at: DATETIME, NULL

Primary Key

- (service_ticket_id, gear_item_id)

c) Entity-Relationship Diagram (ERD)

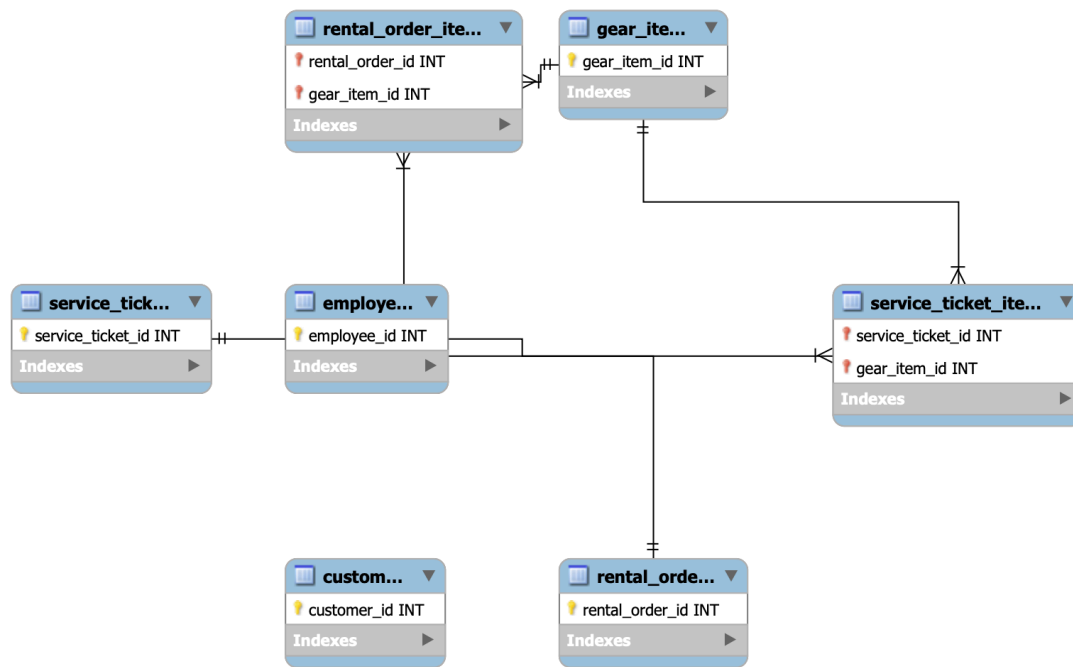


FIGURE 1: ERD

d) Schema Diagram + Normalization (Write-up)

The following report was generated by joining rental orders, customers, employees, and gear items into a single flattened result set, where each row represents a gear item within a rental order. This report reflects how the data might appear in a spreadsheet prior to normalization.

	rental_order_id	order_created_at	customer_name	customer_phone	customer_email	employee_name	employee_role	
1	1	2026-02-01 10:00:00	John Doe	541-555-0101	john.doe@example.com	Shaun White	rental_associate	
2	1	2026-02-01 10:00:00	John Doe	541-555-0101	john.doe@example.com	Shaun White	rental_associate	
3	1	2026-02-01 10:00:00	John Doe	541-555-0101	john.doe@example.com	Shaun White	rental_associate	
4	2	2026-02-02 11:15:00	Noah Patel	<null>	noah.patel@example.com	Shaun White	rental_associate	
5	3	2026-02-03 09:20:00	Mia Johnson	541-555-0103	<null>	Shaun White	rental_associate	
	serial_number	category	brand	model	size	checked_out_at	due_at	returned_at
1	SNB-0001	snowboard	Burton	Process	155	2026-02-01 10:05:00	2026-02-03 10:05:00	<null>
2	BTS-0101	boots	K2	Maysis	10	2026-02-01 10:05:00	2026-02-03 10:05:00	<null>
3	HLT-0201	helmet	Smith	Vantage	M	2026-02-01 10:05:00	2026-02-03 10:05:00	<null>
4	SNB-0001	snowboard	Burton	Process	155	2026-02-02 11:20:00	2026-02-04 11:20:00	2026-02-03 16:45:00
5	SNB-0001	snowboard	Burton	Process	155	2026-02-03 09:25:00	2026-02-05 09:25:00	<null>

FIGURE 2: UNNORMALIZED RENTAL ACTIVITY SAMPLE REPORT DEMONSTRATING DATA REDUNDANCY AND UPDATE/INSERT/DELETE ANOMALIES PRIOR TO NORMALIZATION.

In the sample report, order-level, customer, and employee attributes are repeated across multiple rows because each row represents a single gear item within a rental order. For example, rental order 1 appears three times, resulting in duplicated customer and employee data. This redundancy introduces update anomalies (e.g., a change to a customer's contact information must be applied to multiple rows), insert anomalies (new customers, gear items, or employees cannot be added independently without creating a full rental transaction), and delete anomalies (removing the final line item for an order may unintentionally remove associated order or customer information).

To convert the design to First Normal Form (1NF), repeating groups were eliminated by separating rental order data from individual gear items. Each rental order may contain multiple gear items, so a separate intersection table was introduced to represent one row per gear item per order. The composite key (`rental_order_id`, `gear_item_id`) uniquely identifies each row. With atomic attributes and no repeating groups, the design satisfies 1NF.

In the 1NF design, partial dependencies existed because order-level attributes depended only on `rental_order_id`, while gear attributes depended only on `gear_item_id`. To convert the design to Second Normal Form (2NF), these partial dependencies were removed by decomposing the schema into `rental_orders`, `gear_items`, and `rental_order_items`. Each non-key attribute now depends on the entire primary key of its table.

Transitive dependencies were identified where customer and employee attributes depended on non-key identifiers in rental orders (e.g., `rental_order_id` → `customer_id` → customer attributes). These dependencies were eliminated by moving customer information into a customers table and employee information into an employees table, referenced by foreign keys. After this decomposition, all non-key attributes depend only on the primary key of their respective tables, and the schema satisfies Third Normal Form (3NF).

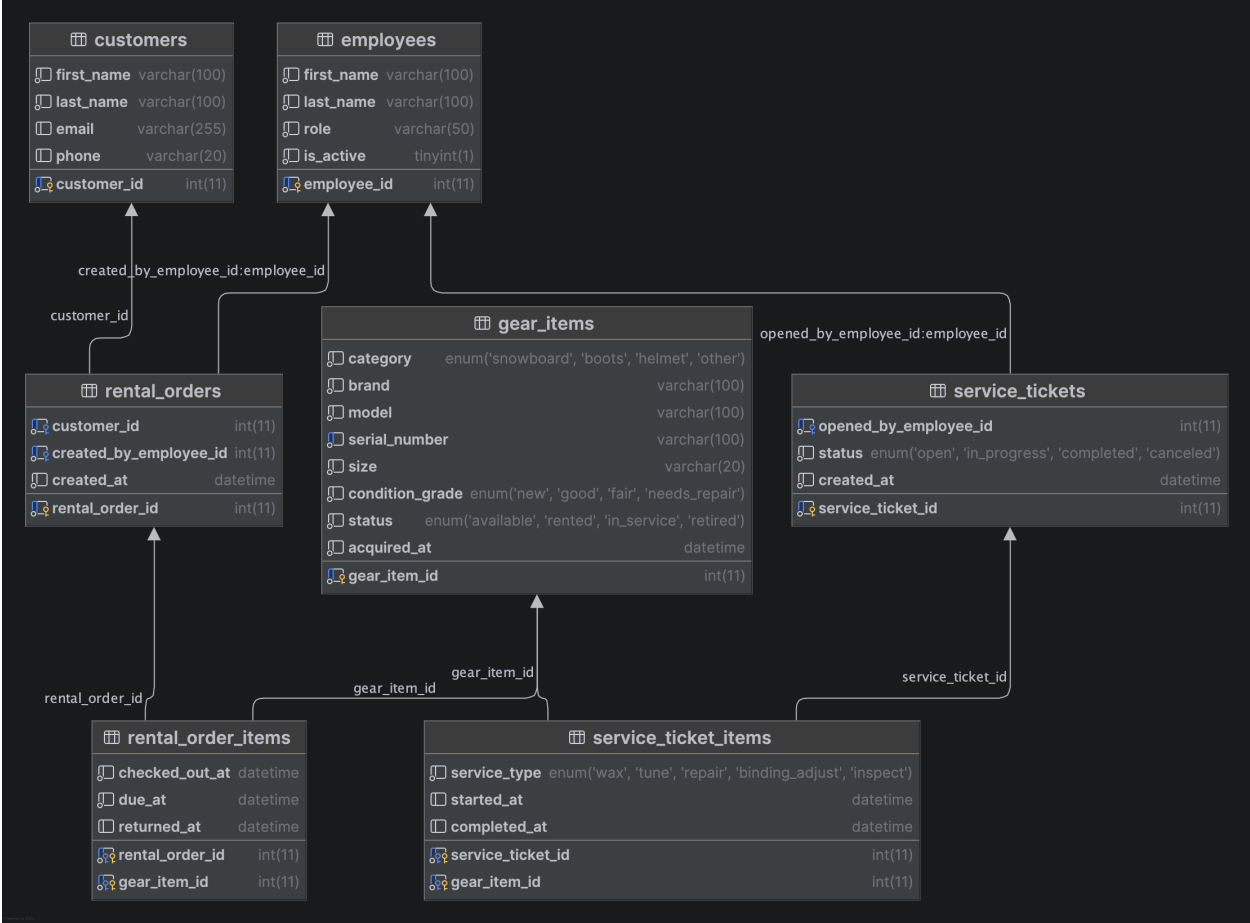


FIGURE 3: FINAL NORMALIZED DATABASE SCHEMA FOR THE SKI RESORT GEAR RENTAL SYSTEM.

e) Example Data

customers

	customer_id	first_name	last_name	email	phone
1	1	John	Doe	john.doe@example.com	541-555-0101
2	2	Noah	Patel	noah.patel@example.com	<null>
3	3	Mia	Johnson	<null>	541-555-0103

employees

gear_items

	employee_id	first_name	last_name	role	is_active
1	1	Shaun	White	rental_associate	1
2	2	Travis	Rice	service_tech	1
3	3	Lindsey	Vonn	manager	0

	gear_item_id	category	brand	model	serial_number	size	condition_grade	status	acquired_at
1	1	snowboard	Burton	Process	SNB-0001	155	good	available	2024-11-01 09:00:00
2	2	boots	K2	Maysis	BTS-0101	10	fair	rented	2023-12-10 10:30:00
3	3	helmet	Smith	Vantage	HLT-0201	M	new	in_service	2024-10-15 14:00:00
4	4	other	Dakine	Wrist Guard	OTH-0301	L	good	retired	2022-01-05 08:15:00

rental_orders

	rental_order_id	customer_id	created_by_employee_id	created_at
1	1	1	1	2026-02-01 10:00:00
2	2	2	2	2026-02-02 11:15:00
3	3	3	3	2026-02-03 09:20:00

rental_order_items

	rental_order_id	gear_item_id	checked_out_at	due_at	returned_at
1	1	1	2026-02-01 10:05:00	2026-02-03 10:05:00	<null>
2	1	1	2026-02-01 10:05:00	2026-02-03 10:05:00	<null>
3	1	1	2026-02-01 10:05:00	2026-02-03 10:05:00	<null>
4	2	2	2026-02-02 11:20:00	2026-02-04 11:20:00	2026-02-03 16:45:00
5	3	3	2026-02-03 09:25:00	2026-02-05 09:25:00	<null>

service_ticket_items

	service_ticket_id	gear_item_id	service_type	started_at	completed_at
1	1	3	inspect	<null>	<null>
2	2	1	binding_adjust	2026-02-02 13:25:00	<null>
3	2	3	repair	2026-02-02 13:15:00	<null>
4	3	3	wax	2026-02-03 15:20:00	2026-02-03 16:05:00

service_tickets

	service_ticket_id	opened_by_employee_id	status	created_at
1	1	2	open	2026-02-01 08:30:00
2	2	2	in_progress	2026-02-02 13:00:00
3	3	3	completed	2026-02-03 15:10:00